

Suheum Heo

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EDUCATION

University of Wisconsin Madison, Madison, WI

May 2027

Bachelor of Science in Computer Science and Data Science

Relevant Coursework: Algorithms, Machine Learning, Database Management Systems, Theory & Algorithms for Data Science

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, R, HTML/CSS

Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, DuckDB, Jupyter Notebook

Frameworks: React, Next.js, Node.js, FastAPI, Django/DRF, Flask, Streamlit, Tailwind CSS

Databases & Tools: PostgreSQL, Supabase, Redis, Docker, Prisma, Git/GitHub, Vercel

AI & APIs: Anthropic Claude API, REST APIs, prompt engineering

PROJECTS

ScoutEdge — Football Transfer Intelligence Platform | *Next.js, TypeScript, Tailwind CSS, Anthropic Claude API, Vercel*

- Built a full-stack AI analytics platform using Next.js, TypeScript, and Anthropic Claude to combine live sports data, structured validation, and LLM reasoning into transfer recommendations and squad analysis
- Designed a multi-source data pipeline merging football-data.org, API-Football, FotMob, and Transfermarkt to separate verified player/club data from LLM-generated tactical reasoning, with hallucination prevention and fact-grounding guardrails

Madison Insights | *Python, DuckDB, Streamlit, Parquet, Folium*

- Built a dual-source ETL pipeline ingesting 73,691 WI State Patrol crash records from historical ZIP/CSV files and WisTransPortal REST APIs using a concurrent Python fetcher (20 workers)
- Designed a severity-weighted SQL hotspot ranking ($\text{fatal} \times 5$, $\text{injury} \times 2$, $\text{PDO} \times 1$) to surface genuinely dangerous intersections; identified S Stoughton Rd @ Pflaum Rd as Dane County's #1 hazard with E Washington Ave appearing 4 $\times$ in the top 15
- Migrated from PostgreSQL to DuckDB + Parquet for zero-cost, zero-secret Streamlit Cloud deployment; implemented pure-Python ray casting spatial join mapping 142 Census tracts to Madison neighborhoods with no GeoPandas dependency

FootyLens — Football Analytics Platform | *FastAPI, Django/DRF, PostgreSQL, Redis, Next.js, Docker*

- Architected a 5-service system containerized with Docker Compose; designed strict service boundary separating async data ingestion (FastAPI) from ORM/auth (Django + DRF), with TypeScript interfaces mirroring Pydantic schemas to catch mismatches at compile time
- Built idempotent bulk-sync eliminating 760+ DB round-trips per run and a token-bucket rate limiter with Redis TTL cache reducing live API calls to near-zero under normal load
- Delivered analytics endpoints for xG proxy, last-N form, and top scorers using pandas/NumPy computation with Pydantic v2 validation and nullable-field sanitization; surfaced as a server-rendered Next.js dashboard

MoneyMap — Personal Finance App | *Next.js 14, TypeScript, Supabase, Tailwind CSS, Chart.js*

- Built a full-stack PWA tracking 500+ real expense entries across currencies, categories, and locations with live exchange rate integration, Supabase RLS multi-user isolation, and Google OAuth
- Deployed on Vercel with installable iPhone PWA support, 7-language i18n via i18next, and locale-aware date/number formatting

Common Ground — AI Classroom Discussion Tool | *Node.js, Express, PostgreSQL, Anthropic Claude API*

- Built a real-time classroom tool using Claude to surface shared values and genuine tensions across student perspectives; session-based architecture with QR code joining and live result broadcasting
- Selected as finalist in Anthropic $\times$ CDIS Speed-Hackathon

WORK EXPERIENCE

Culinary Assistant & Shift Lead, Liz's Market, UW–Madison, Madison, WI

Nov 2024 – Jul 2025

- Supervised 5–8 student staff in a high-volume dining facility serving 500+ meals daily; trained 10+ new employees on safety protocols and workflow optimization

Cashier, Four Lakes Market, UW–Madison, Madison, WI

Dec 2025 – Present

- Handled 200+ daily transactions in a fast-paced university dining environment

Military Service — Basic Training & Social Service Agent, Korea Army, Ulsan, South Korea

Oct 2022 – Jul 2024

- Provided care for elderly residents with dementia; collaborated with staff across diverse age groups in a structured elder-care environment